MEDICAL ROOM AUTOMATION SYSTEM

MRAS v3.0 — Intelligent Predictive Health Platform

Project Proposal

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Institution	Faculty of Information Technology, University of Vocational Technology
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Student No.	Name	Role
SIS/24/B2/36	L.B. Charith Jeewan	Team Lead
SIS/24/B2/38	W.I.L. Withana	Architect
SIS/24/B2/13	H.K.G.V. Lakmali Koralage	Developer
SIS/24/B2/15	G.B.D. Anuradha	Security Lead
SIS/24/B2/39	B.W.S.S. Nawarathna	DevOps Lead

1. Executive Summary

The Medical Room Automation System version 3.0 (MRAS v3.0) is a next-generation intelligent health platform designed to transform corporate medical facilities from reactive treatment centres into proactive, predictive wellness ecosystems. Building upon the foundational MRAS v2 architecture — which addressed digitisation of patient records, pharmacy management, and reporting — this proposal introduces three breakthrough innovations that no existing corporate medical room system worldwide has fully addressed.

MRAS v3.0 introduces: (1) a Multi-Source Predictive Health Forecasting Engine that fuses user-input data, mobile device sensor streams, and real-time environmental climate APIs to predict illness before symptoms emerge; (2) a Job-Role Intensity and Social Support Index (JRISSI) — a proprietary mental health risk scoring model designed specifically for corporate occupational contexts; and (3) a Closed-Loop Smart Notification and Self-Behaviour Correction Engine that delivers personalised, AI-generated health interventions directly to employees and corporate doctors, closing the gap between data collection and actionable care.

These innovations are not incremental improvements. They represent a paradigm shift: from a system that records what happened to a system that prevents what could happen. This document presents the worldwide gaps in current MRAS solutions, the precise innovations proposed to fill each gap, and a rigorous justification of their clinical, operational, and academic significance.

2. Worldwide Gaps in Medical Room Automation Systems

A comprehensive review of global medical room automation deployments in 2025 — including systems deployed across the United States, United Kingdom, European Union, India, Singapore, and Australia — reveals eight critical, recurring gaps. These gaps are documented across academic literature, industry reports (WHO, WEF, Mordor Intelligence, HIMSS), and clinical case studies. Each gap represents an unmet need that MRAS v3.0 is specifically architected to address.

#	Worldwide Gap Identified	Global Impact	MRAS v3.0 Solution	Innovation Level
G 1	Reactive-only care: systems record illness after a patient presents; no system predicts illness before it occurs	72–99% of hospital alerts are non-action able because they trigger only on crisis events (HIMSS, 2025)	Multi-Source Predictive Forecasting Engine fusing biometric, behavioural, and environmental data to predict illness 3–14 days in advance	**World-first for corporate medical rooms**
G 2	No environmental disease forecasting: weather and climate data are never connected to health systems despite well-documented links between humidity, temperature, and respiratory/allergic illness	Respirator y diseases spike 40% during high-humi dity seasons in South Asia yet no system anticipates this (WHO, 2024)	Real-time climate API integration predicts asthma, allergy, and heat-stress episodes per employee based on their medical history and current weather conditions	**Novel — no comparable corporate implementation exists**
G 3	Mental health is entirely absent from medical room systems: no existing MRAS anywhere in the world captures occupational mental health risk or workplace stress indicators	Mental disorders affect 1 billion people globally; corporate health systems miss 90% of early-stag e cases (PMC, 2025)	JRISSI (Job-Role Intensity and Social Support Index) — a proprietary algorithm scoring mental health risk from job role, workload signals, and social support patterns	**Original algorithm, academically novel**
G 4	No personalised self-behaviour correction: systems provide generic	Generic wellness tips have <15%	AI engine generates hyper-personalised behaviour corrections (e.g. 'Reduce sugar	**Clinically significant innovation**

	health advice, not situation-specific, AI-generated interventions tied to an individual's real-time health trajectory	engagement rate vs 67% for personalised digital interventions (Journal of Medical Internet Research, 2024)	intake over next 3 days — your glucose trend indicates pre-diabetic risk') triggered by predictive model outputs	
G5	Longitudinal health tracking is absent: most systems store isolated visit records with no trend analysis over time	Doctors make decisions without access to 6–12 month health trends, increasing misdiagnosis risk by 35% (The Lancet Digital Health, 2024)	Automated longitudinal health profiling analyses blood pressure, weight, activity, and consultation patterns over time to surface emerging chronic disease risk	**Core to the MRAS v3.0 data architecture**
G6	No pre-visit doctor briefing: doctors receive no patient summary before the patient enters the room	Doctors spend 3–5 minutes per consultation retrieving and reviewing records — 30–40% of total consultation time (Duke University, 2024)	Automated digital health summary pushed to doctor's dashboard the moment patient scans in via Employee ID — includes health risk score, active medications, allergies, and AI-flagged concerns	**Closes a critical workflow gap**
G7	Mobile and wearable data streams are ignored: corporate health systems do not capture physical activity, step counts, or mobility patterns despite broad smartphone availability	McKinsey (2025): 10–20% of clinical shift time is spent on tasks that mobile-enabled automation could eliminate	Web Bluetooth and device motion APIs capture physical activity from employee smartphones — no specialist hardware required — feeding directly into the predictive health model	**Hardware-free IoT integration — highly feasible**

G8	Closed-loop feedback is missing: no system automatically monitors whether a health intervention was followed, adapted to outcomes, or escalated	Without closed-loop feedback, 68% of digital health interventions produce no measurable behaviour change (BMJ Digital Health, 2024)	Smart Notification Engine tracks intervention outcomes, adapts recommendations if compliance is low, and automatically escalates to the doctor if a high-risk trend persists	**Closes the intervention-to-outcome loop**

3. Proposed System — MRAS v3.0 Architecture

3.1 System Overview

MRAS v3.0 extends the proven Modular Monolith architecture of MRAS v2 with three new intelligence layers that operate above the existing five core modules. These layers — the Data Fusion Layer, the Predictive Intelligence Layer, and the Smart Notification Layer — are designed to be cleanly separable from the core system, preserving the architectural integrity of the Modular Monolith while introducing AI capabilities that can be independently developed, tested, and deployed.

3.2 The Three-Stream Data Architecture

MRAS v3.0 captures continuous health intelligence from three primary data streams, each addressing a distinct dimension of employee health.

Stream 1 — User Input Layer

Employees interact with a structured daily health check-in interface (accessible via their employee portal or mobile browser) that captures: personal dietary intake and meal patterns; self-reported stress levels and sleep quality; physical symptoms and medication compliance; and historical medical context from previous consultations. This layer transforms the MRAS from a passive record system to an active, participatory health platform.

Stream 2 — Device and Sensor Layer

Using the Web Device Motion API and Web Bluetooth API — both accessible from standard mobile browsers without requiring specialist hardware installation — MRAS v3.0 captures physical activity metrics including step count, walking distance, movement intensity, and inactivity duration. For employees who opt into wearable integration (Fitbit, Apple Watch, Samsung Health via their respective open APIs), richer biometric streams including heart rate variability, sleep stages, and blood oxygen saturation are available. Critically, this integration requires no hospital-grade hardware investment — a smartphone is sufficient.

Stream 3 — Environmental Layer

A scheduled background service queries public climate APIs (OpenWeatherMap API — free tier; Tomorrow.io API) for the organisation's geographic location. The retrieved data — temperature, humidity, air quality index (AQI), UV index, and pollen count — is stored against a daily timestamp and correlated with the employee population's historical illness patterns. This enables the system to forecast disease risk for the coming days based on predicted environmental conditions, a capability no current corporate medical room system possesses.

3.3 The Predictive Intelligence Layer

3.3.1 Physical Health Forecasting Engine

The Physical Health Forecasting Engine applies time-series machine learning models (Facebook Prophet for seasonal forecasting; scikit-learn Random Forest for risk classification) to the three data streams. Its primary outputs are:

- Environmental Disease Forecast: predicts respiratory illness risk (asthma, allergic rhinitis) 3–7 days in advance based on forecast humidity and AQI levels correlated with each employee's medical history of respiratory conditions.
- Chronic Disease Risk Trajectory: analyses 3–12 month trends in blood pressure, weight, glucose readings (from consultation records), and physical activity to generate a risk score for hypertension, Type 2 diabetes, and obesity-related conditions.
- Heat-Stress Alert: identifies employees at risk of heat-related illness during high-temperature periods based on their age, BMI, and physical activity levels, issuing preventive alerts before any clinical event occurs.

Clinical Evidence	AI monitoring systems integrating biometric sensor data and environmental signals predicted symptom exacerbation with 89% accuracy in 2025 trials (PMC/Nature, 2025). Lee et al. demonstrated 91% accuracy in predicting depressive episodes using wearable + mobile data, underscoring the validity of multi-stream data fusion for health forecasting.

3.3.2 Job-Role Intensity and Social Support Index (JRISSI)

The JRISSI is an original algorithm developed by Group 03 to address Gap G3 — the complete absence of occupational mental health scoring in any current MRAS worldwide. The algorithm computes a composite Mental Health Risk Score (MHRS) on a 0–100 scale by weighting four subscores:

JRISSI Sub-Score	Data Sources	Weight	Rationale
Job Role Intensity (JRI)	Role classification, workload self-report, overtime frequency from HRIS	**30%**	High-intensity roles (management, emergency response) carry 3× burnout risk
Social Support Index (SSI)	Self-reported inner circle size, frequency of social interaction, remote/on-site status	**25%**	Social isolation is the strongest predictor of workplace depression (SHRM, 2025)
Physical Health Indicators	BMI trend, sleep quality score, physical activity from Stream 2	**25%**	Bidirectional relationship between physical and mental health is well established
Consultation Pattern Score	Frequency of stress-related consultations, medication types, recurrence rate	**20%**	Longitudinal consultation pattern is the strongest objective mental health signal in this dataset

The MHRS output — Low (0–39), Moderate (40–69), High (70–100) — is displayed on the doctor's dashboard as a mental health risk badge and triggers the Smart Notification Engine if a Moderate or High score is sustained for more than 14 days. This is the first occupational mental health scoring algorithm designed for integration within a corporate medical room system.

Research Validation	Explainable AI (XAI) research using the OSMI Mental Health dataset (2016–2023) found that workplace-specific features — role type, employer importance of mental health, and company size — are among the top predictors of workplace mental health outcomes (MDPI Informatics, November 2025). The JRISSI directly operationalises these research findings in a corporate health context.

3.4 Predictive Health Forecasting and Self-Behaviour Correction

When the Predictive Intelligence Layer identifies a risk — whether physical (e.g. rising blood pressure trend) or mental (MHRS ≥ 40) — the system does not simply generate a clinical alert. It generates a personalised, AI-crafted self-behaviour correction: a specific, time-bounded, actionable recommendation delivered to the employee via their preferred notification channel.

Examples of system-generated behaviour corrections:

Trigger Condition	Risk Identified	AI-Generated Behaviour Correction
BP readings rising over 3 weeks + low activity	**Pre-hypertension risk**	"Your blood pressure trend suggests developing hypertension risk. Recommended: 20-minute walk daily for 5 days, reduce sodium intake this week. Doctor notified."
High humidity forecast + history of asthma	**Respiratory episode in 48h**	"High humidity forecast for the next 3 days. Given your respiratory history, consider keeping your inhaler accessible and avoiding outdoor activity between 10am–2pm."
MHRS = 72, sustained 18 days, high JRI role	**High mental health risk**	"Your mental wellness score indicates elevated stress. A 15-minute mindfulness session is suggested today. Your corporate doctor has been notified and will reach out."
Step count <2,000/day for 7 consecutive days	**Sedentary behaviour risk**	"You have been predominantly sedentary this week. Suggested: take a 10-minute walking break at 11am and 3pm today. Hydration reminder: 8 glasses of water by 6pm."

3.5 Smart Medical Reporting and Doctor Notification

3.5.1 Longitudinal Health Profile

Every employee has a continuously updated Longitudinal Health Profile — a structured data object maintained in PostgreSQL that aggregates all three data streams over time. The profile is the input to all predictive models and is surfaced to the doctor as an interactive timeline dashboard. Unlike traditional medical records, which present isolated visit snapshots, the Longitudinal Health Profile surfaces patterns: a doctor can see that an employee's BMI has increased 8% over six months, their average step count has halved, and their MHRS has risen from 32 to 68 — none of which would be visible from a single consultation record.

3.5.2 Automated Doctor Briefing (Pre-Visit Push)

When an employee scans their QR code or enters their Employee ID at the medical room reception, the system instantly pushes a structured health summary to the attending doctor's dashboard. This summary includes: current health risk scores (physical and mental); active medications, allergies, and drug-interaction flags; AI-flagged concerns from the predictive engine; and the three most significant trends from their Longitudinal Health Profile. The doctor enters the consultation fully briefed, reducing administrative overhead by an estimated 3–5 minutes per patient — consistent with the 20% note-taking time reduction observed in AI-assisted clinical documentation studies (Duke University, 2024).

3.5.3 Critical Escalation Alerts

The system monitors all employees' predictive risk scores continuously. If a High-risk classification is sustained for more than 14 days without a recorded consultation, the system automatically generates a Critical Escalation Alert to the corporate doctor — not a generic notification, but a structured clinical summary containing the employee's ID, the specific risk indicators driving the alert, and the recommended action. This proactive escalation mechanism directly addresses the global gap in closed-loop health intervention (Gap G8).

4. Innovation Justification

4.1 Global Novelty Assessment

The following table rates each MRAS v3.0 innovation against a ten-point novelty scale based on a review of existing commercial systems (Epic, Cerner, SAP Health, HealthConnect, Oracle Health), academic literature (PubMed, IEEE, Springer), and industry reports (HIMSS 2025, Mordor Intelligence, WEF). A score of 10 indicates no comparable implementation exists worldwide; a score of 7 indicates partial implementations exist but lack the integration proposed here.

Innovation	Score	Novelty Justification
Environmental Disease Forecasting linked to individual medical history	10 / 10	No corporate medical room system worldwide integrates climate API data with individual patient health history to produce personalised disease forecasts. Academic literature confirms the link between humidity and respiratory illness but no deployed system operationalises this at individual level.
JRISSI Mental Health Risk Algorithm for occupational settings	10 / 10	No existing MRAS product includes a mental health risk scoring engine. The JRISSI is an original algorithm designed specifically for the corporate occupational context, combining job role intensity with social support signals and physical health indicators. This is an original academic and technical contribution.
Multi-stream data fusion (user input + mobile sensors + climate) for prediction	9 / 10	Academic research confirms the value of multimodal data fusion for health prediction (PMC, 2025: 91% accuracy). No corporate health system has deployed this combination. The hardware-free approach using Web APIs is a further novel contribution — comparable systems require dedicated IoT infrastructure.
AI-Generated Personalised Self-Behaviour Correction Engine	8 / 10	Personalised digital interventions achieve 67% engagement vs 15% for generic advice (JMIR, 2024). While consumer wellness apps provide personalised tips, no corporate medical room system generates contextual, clinically-triggered, time-bounded behaviour corrections with physician oversight.
Automated Pre-Visit Doctor Briefing Push Notification	7 / 10	Ambient documentation research (Duke, Mass General Brigham) demonstrates significant time savings from AI-assisted pre-briefing. The specific implementation — triggered by Employee ID scan and delivered before the patient enters the room — is novel in the corporate medical room context.
Closed-Loop Health Intervention Monitoring (escalation if ignored)	8 / 10	Closed-loop feedback is identified as a critical gap across all global healthcare automation reviews (BMJ Digital Health, 2024). MRAS v3.0's automatic escalation to the doctor when an employee's risk score remains high without a consultation response directly closes this gap.
Hardware-Free Physical Activity Capture via Web Browser APIs	8 / 10	Using the Web Device Motion API and Web Bluetooth API to capture physical activity without deploying any specialist sensors is a technically novel approach that eliminates the primary cost barrier to IoT health monitoring in corporate settings.

4.2 Comparison with Existing Systems

The table below compares MRAS v3.0 against the four most commonly deployed corporate health information systems worldwide, demonstrating the scope of the innovation gap that MRAS v3.0 addresses.

Capability	Epic (USA)	Cerner (Global)	SAP Health	HealthConnect	MRAS v3.0
Digital patient records	✓	✓	✓	✓	✓
Pharmacy / inventory management	✓	✓	✓	Partial	**✓ (FEFO AI)**
Environmental disease forecasting	✗	✗	✗	✗	**✓ World-first**
Mental health risk scoring	✗	✗	✗	✗	**✓ JRISSI Algorithm**
Mobile sensor data integration	✗	✗	Partial	✗	**✓ No hardware needed**
Personalised behaviour correction	✗	✗	✗	✗	**✓ AI-generated, contextual**
Longitudinal health trend analysis	Partial	Partial	✗	✗	**✓ Automated profiling**
Pre-visit doctor briefing push	✗	Partial	✗	✗	**✓ Automated, real-time**
Closed-loop escalation if risk persists	✗	✗	✗	✗	**✓ Novel feature**
Suitability for corporate medical rooms	Hospital-scale only	Hospital-scale only	Enterprise only	✓	**✓ Purpose-built**

5. System Architecture — MRAS v3.0

5.1 Architecture Overview

MRAS v3.0 retains the Modular Monolith architecture validated in v2, adding three new intelligence modules that integrate cleanly with the existing five core modules. All inter-module communication continues to occur via in-memory function calls within the FastAPI backend process, preserving the performance and ACID transactional integrity of the original design.

5.2 Module Structure

Module	Responsibilities	Key Technologies
User & Auth (existing)	Identity, roles, JWT session management	python-jose, passlib/bcrypt, RBAC
Patient Management (existing)	Demographics, Employee ID lookup, medical history	SQLAlchemy 2.0, PostgreSQL 15+
Consultation Module (existing)	Diagnoses, prescriptions, drug-interaction alerts	FastAPI, OpenFDA API, Pydantic v2
Inventory & Pharmacy (existing)	FEFO stock management, GRN processing, expiry alerts	PostgreSQL FEFO function, APScheduler
Reporting & Audit (existing)	Dashboards, audit logs, management reports	Recharts / D3.js, FastAPI, PostgreSQL views
Data Fusion Layer (NEW v3.0)	Ingests Stream 1, 2, 3; normalises and stores time-series health data	Web Device Motion API, Web Bluetooth, OpenWeatherMap API, TimescaleDB extension
Predictive Intelligence Layer (NEW v3.0)	Physical forecasting engine, JRISSI mental health scorer, longitudinal profile builder	scikit-learn, Facebook Prophet, pandas, FastAPI background tasks
Smart Notification Layer (NEW v3.0)	Behaviour correction engine, WhatsApp/email push, doctor briefing, escalation monitor	Twilio API, SendGrid, FastAPI BackgroundTasks, Anthropic Claude API (behaviour text generation)

5.3 Data Flow

The data flow of MRAS v3.0 operates as a continuous pipeline: Stream 1 (user input) + Stream 2 (device sensors) + Stream 3 (climate API) → Data Fusion Layer → time-series storage in PostgreSQL (with TimescaleDB extension for efficient time-series queries) → Predictive Intelligence Layer runs nightly batch jobs and real-time event-triggered analyses → outputs (risk scores, forecasts, JRISSI scores) → Smart Notification Layer dispatches personalised interventions to employees and automated briefings to doctors → outcomes monitored via closed-loop feedback → escalation triggered if risk persists without resolution.

6. Technology Stack — MRAS v3.0

Layer	Technology	Purpose in v3.0
Frontend	React 18 + TypeScript + MUI	Role dashboards, health timeline, JRISSI risk badge, live queue UI
Mobile APIs	Web Device Motion API, Web Bluetooth API	Hardware-free physical activity capture from employee smartphones
Backend	Python 3.12 + FastAPI (async)	REST API, business logic, predictive engine endpoints, RBAC
ML / AI	scikit-learn, Facebook Prophet, pandas	Physical health forecasting, JRISSI scorer, longitudinal trend analysis
LLM Integration	Anthropic Claude API (claude-sonnet-4-6)	Generates personalised self-behaviour correction text; doctor summary narratives
Database	PostgreSQL 15 + TimescaleDB	ACID compliance + efficient time-series queries for longitudinal health data
Climate API	OpenWeatherMap API / Tomorrow.io	Real-time and forecast weather data for environmental disease prediction
Notifications	Twilio WhatsApp API, SendGrid	Patient medication reminders, behaviour corrections, doctor alerts
Containerisation	Docker + Docker Compose	Consistent development and production environments
CI/CD	GitHub Actions	Automated testing, lint, docker build, staging and production deploy
Cloud	AWS (ECS Fargate + RDS + S3)	HIPAA-aligned managed infrastructure, auto-scaling, disaster recovery

7. Public Dataset Strategy

MRAS v3.0 uses a structured public dataset integration strategy to seed, train, and validate its predictive models. All datasets are free, openly licensed, and mapped directly to specific system modules and predictive features.

Dataset	Source	Records	Usage in MRAS v3.0
Heart Disease (Cleveland)	UCI #45	303	Train physical health risk classifier; seed Patient and Consultation modules
Diabetes 130-US Hospitals	UCI #296	101,766	Train longitudinal trend model; populate Reporting dashboard with realistic multi-year data
Drug Reviews (DrugLib)	UCI #461	215,063	Seed medicine catalogue; validate drug-interaction alert engine
WHO Essential Medicines List (23rd, 2023)	WHO	502 drugs	Primary seed data for Inventory module; maps drug names to therapeutic categories
OSMI Mental Health in Tech Dataset (2016–2023)	OSMI	4,218	Validate and calibrate the JRISSI algorithm; benchmark mental health risk scoring accuracy
Healthcare Dataset (Kaggle)	Kaggle	10,000	Seed Patient module with realistic demographic diversity; test consultation workflows
MoH Sri Lanka Health Statistics	MoH LK	Annual	Calibrate environmental forecasting model with locally relevant disease prevalence data

8. Risk Register

#	Risk	Likelihood	Impact	Mitigation
R1	Predictive model accuracy insufficient for clinical use	**Medium**	**High**	All predictions labelled as advisory only; doctor reviews and confirms before action; model accuracy disclosed to users
R2	Employee data privacy breach via expanded sensor data collection	**Low**	**High**	All sensor data opt-in only; AES-256 at rest; TLS 1.3 in transit; explicit consent screen; data minimisation by design
R3	Climate API service outage breaks environmental forecasting	**Medium**	**Low**	Secondary API fallback (Tomorrow.io as backup to OpenWeatherMap); forecasting degrades gracefully with no prediction rather than system failure
R4	JRISSI mental health score stigmatises employees	**Medium**	**High**	Score is confidential — visible only to treating doctor; never visible to HR or management; explicit in privacy policy
R5	Mobile browser API limitations on certain devices reduce sensor data quality	**High**	**Medium**	System operates fully without sensor data; Stream 2 is additive; graceful degradation documented in UI
R6	Staff resistance to adopting expanded health monitoring features	**Medium**	**Medium**	Phased rollout starting with optional features; staff training programme; champion designation in each department
R7	LLM-generated behaviour correction text is clinically inappropriate	**Low**	**High**	All AI-generated text goes through a rules-based safety filter before delivery; doctor can override or suppress any correction; disclaimer displayed to employee

9. Project Roadmap

Phase	Duration	Deliverables
v2 Core	Months 1–3	Auth module, Patient Management, Consultation, FEFO Pharmacy, Reporting dashboard, CI/CD pipeline, PostgreSQL schema, public dataset seeding
v3.0 — Sprint 1	Month 4	Data Fusion Layer: Stream 1 (health check-in UI), Stream 3 (climate API integration), TimescaleDB setup, FEFO heatmap dashboard
v3.0 — Sprint 2	Month 5	Physical Health Forecasting Engine (Prophet + scikit-learn), environmental disease forecast, FEFO AI demand prediction, drug-interaction alert engine
v3.0 — Sprint 3	Month 6	JRISSI Mental Health Algorithm, Stream 2 (mobile device motion API), QR code check-in, pre-visit doctor briefing push notification
v3.0 — Sprint 4	Month 7	Smart Notification Layer, behaviour correction engine (Claude API), WhatsApp reminders, closed-loop escalation monitor, longitudinal health profile UI
UAT & Demo	Month 8	Full system UAT with simulated employee population using public datasets; performance testing; security penetration testing; final documentation

10. Conclusion

MRAS v3.0 represents a genuine leap forward in corporate healthcare technology. Its three core innovations — the Multi-Source Predictive Health Forecasting Engine, the JRISSI Mental Health Risk Algorithm, and the Closed-Loop Smart Notification Engine — collectively address eight documented global gaps that no current corporate medical room system has resolved. These are not incremental enhancements to existing functionality. They are original technical and clinical contributions that shift the paradigm of occupational health management from reactive documentation to proactive, personalised prevention.

The system is grounded in peer-reviewed evidence: the 91% accuracy of multimodal AI in health prediction (PMC, 2025), the 67% engagement advantage of personalised digital health interventions (JMIR, 2024), the validated relationship between job role intensity and burnout risk (SHRM, 2025), and the documented 20–30% clinical time savings from AI-assisted documentation (Duke University, 2024). Every architectural decision — the retention of the proven Modular Monolith, the use of public datasets for model training, the hardware-free mobile API approach — is justified by both technical rigour and practical feasibility within the project's scope and timeline.

MRAS v3.0 is not an aspirational research prototype. It is a buildable, deployable, academically sound system that Group 03 is fully committed to delivering. We recommend proceeding with the phased roadmap outlined in Section 9, commencing immediately with the MRAS v2 Core sprint.

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